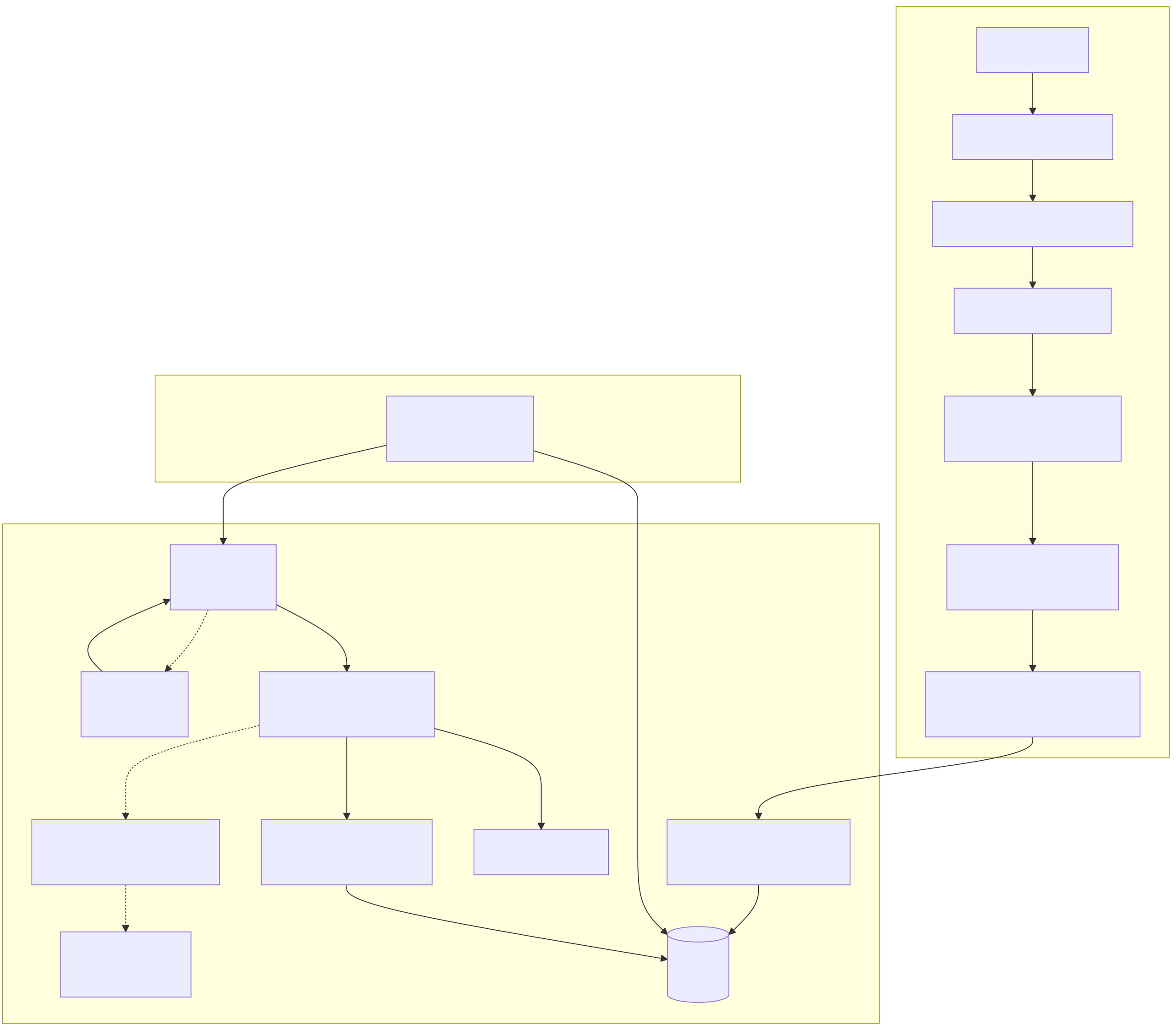


Smart Listings - detailed full architecture

OFFLINE preparation - author machine only

otodom.pl
listing pages + payloads

pipeline/scrape.ts
nationwide random page sampling
balanced 54 sale + 54 rent

data/raw/otodom/*.json
raw snapshots

pipeline/ingest.ts
batch enrichment orchestrator

pipeline/normalize.ts
deterministic parsing
price, area, rooms, floor, location

claude.ts
extractFromDescription
summarizeListing

RUNTIME application - browse works even without API key

Vue 3 SPA
main.ts + App.vue

router.ts
list + detail routes

api/client.ts
typed fetch wrappers

ListingsView.vue
manual filters + AI mode
URL as state, one filters object

ChatSearch.vue
NL query box
POST parse-intent

Express 5 - index.ts
JSON middleware
/api routing + SPA fallback
serveStatic in single-port mode

routes/
listings.ts
search.ts

Zod
boundary
validation

DATA layer

pipeline/seed.ts
upsert committed JSON
flag fuzzy duplicates

db.ts
createPool
ensureSchema() on boot

MySQL 8 - single listings table
columns for normalized listing + raw_json kept internal
indexes: offer_type, city, price, area_m2, dedupe_hash
UNIQUE (source, source_id)
collation utf8mb4_0900_ai_ci for case- and accent-insensitive search

BOOT / DEPLOY / PROCESS TOPOLOGY

docker compose up
db + app
localhost:3004

app container / process
npm run seed && server start
Express serves API + built SPA

db container
mysql:8.4
healthcheck + volume

host-side dev mode
npm run dev
server tsx watch + Vite proxy

single-port prod/demo
browser hits one app port
API and SPA served together

optional AI dependency
missing ANTHROPIC_API_KEY disables chat semantics
browse/search still works

Cross-cutting rules and external integrations

AI confinement
Only claude.ts imports the Anthropic SDK.
Deterministic code owns price, area, rooms, and floor parsing.

Strict layering
routes validate and shape HTTP responses.
repo owns all SQL; routes never inline SQL.

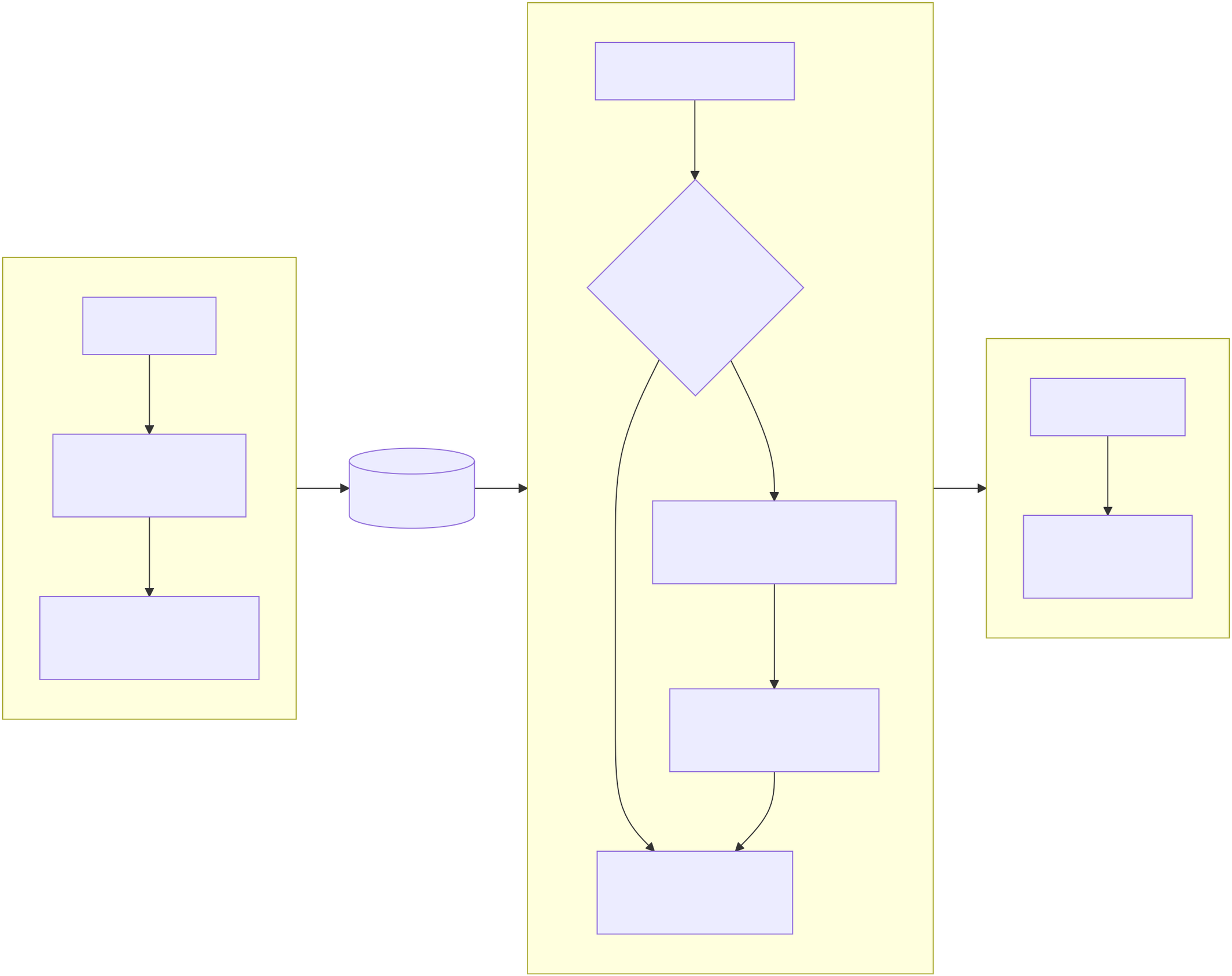
Nulls end to end
Unknown values stay null in pipeline, DB, API, and UI.
Missing data renders as "-", never fake zeroes.

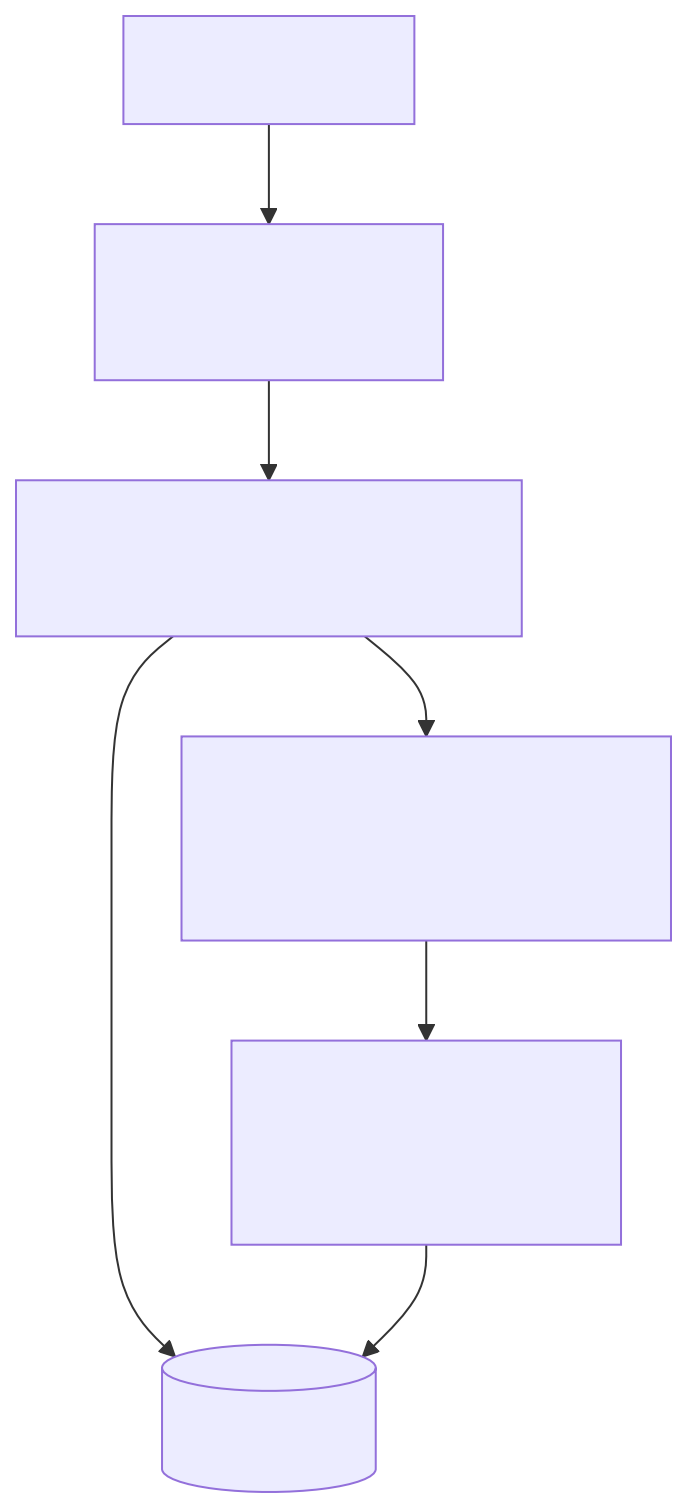
Shareable state
ListingsView syncs filters to URL.
AI mode also persists through mode=ai in the query string.

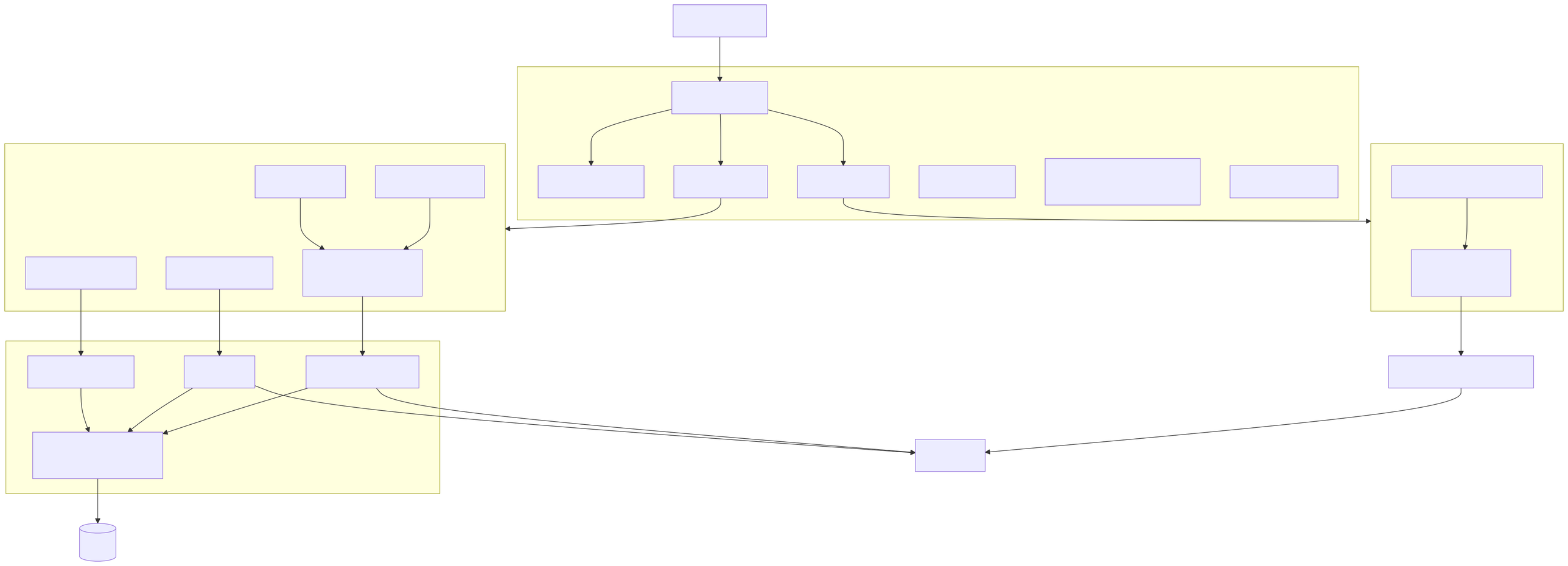
Runtime request flow
Browser → Vue views/components → api/client.ts → Express → routes → Zod → repo → MySQL

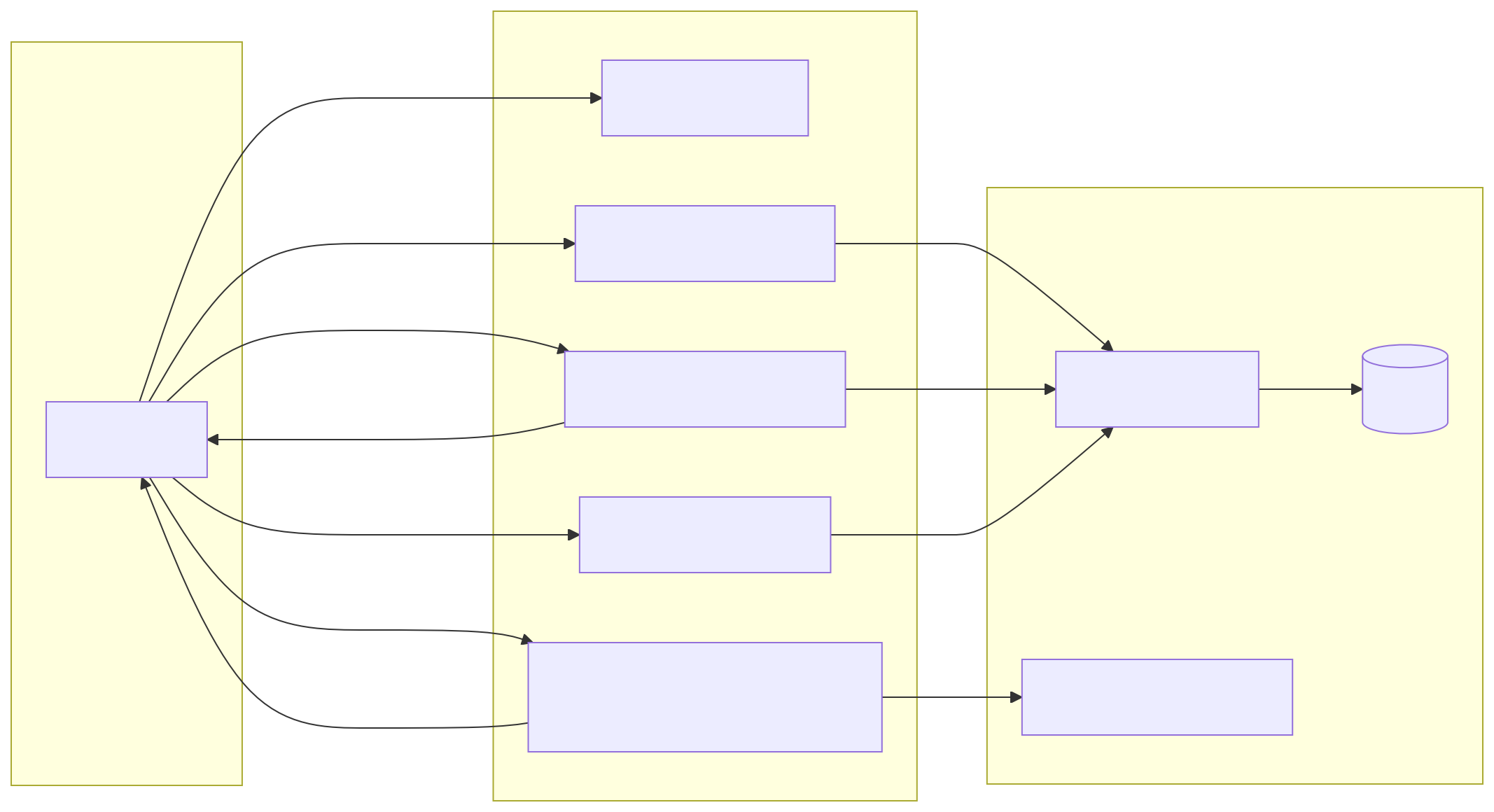
AI request flow
ChatSearch → /api/search/parse-intent → claude.ts → optional Anthropic API → filters JSON → same /api/listings path

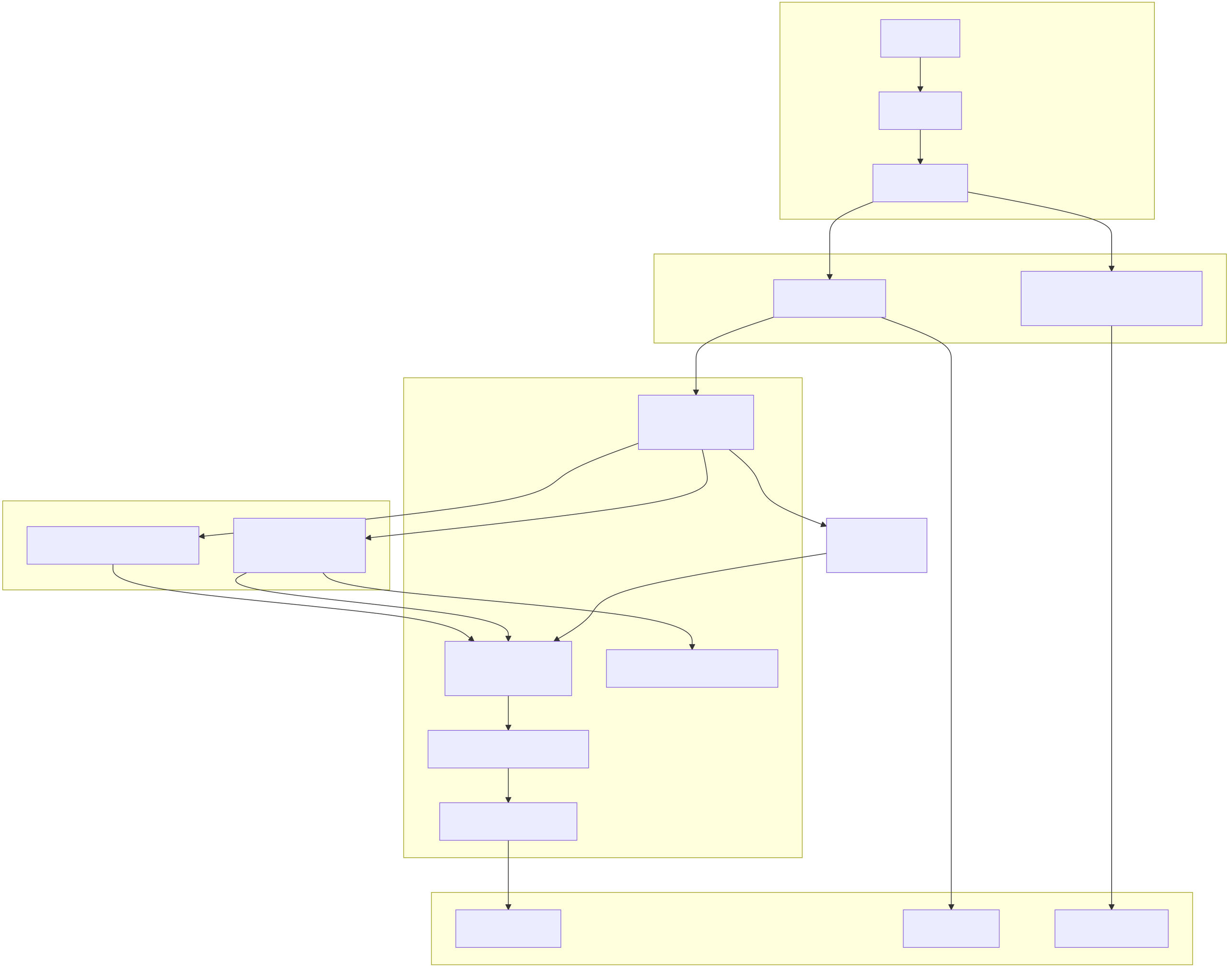
Data prep flow
scrape → raw JSON → ingest → normalize → enrich → listi

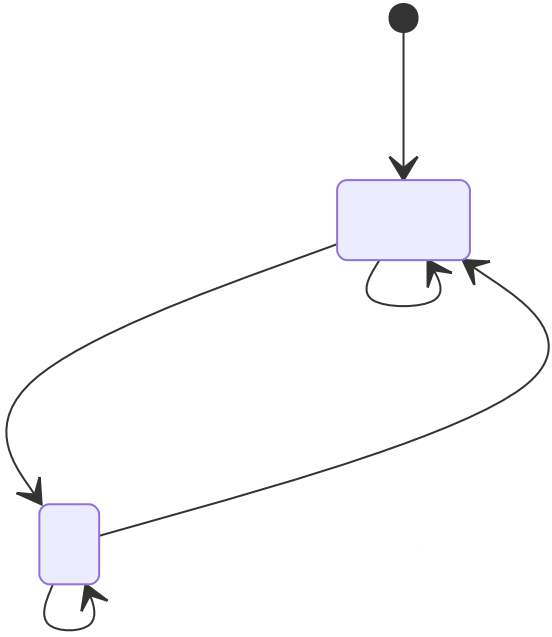


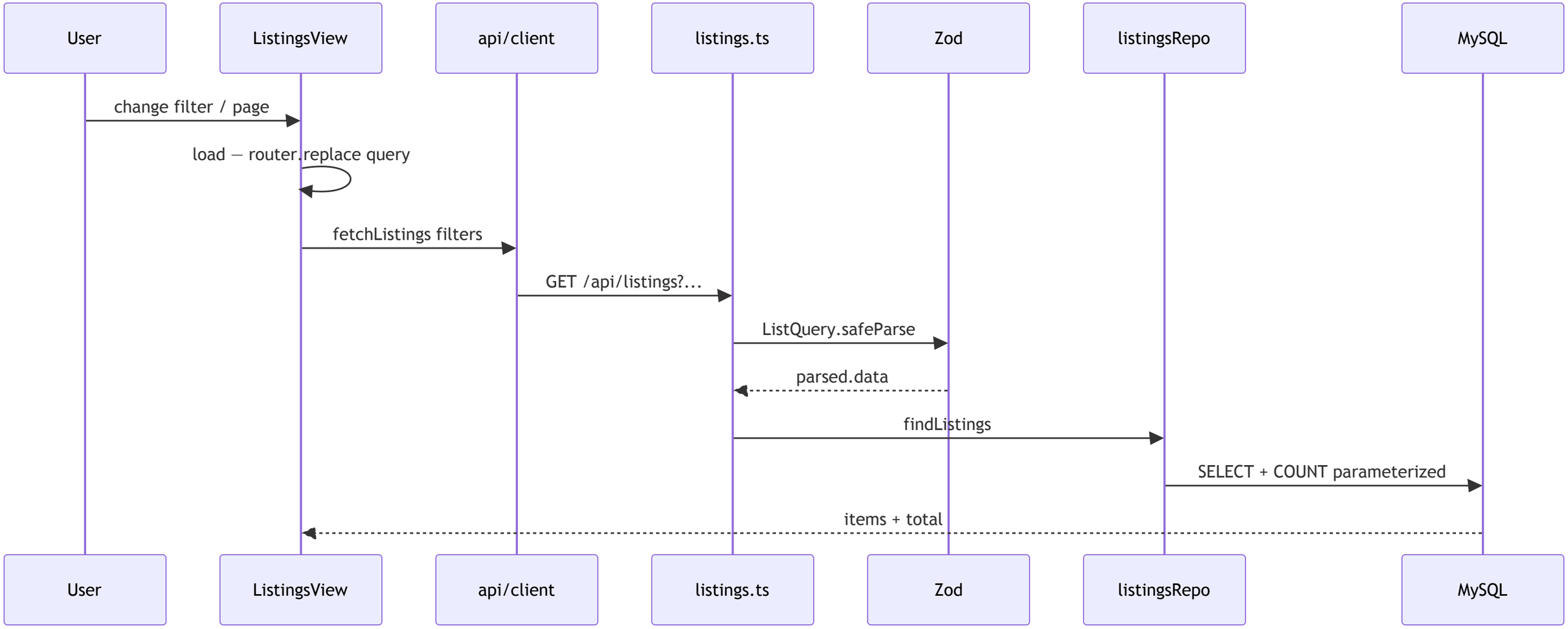


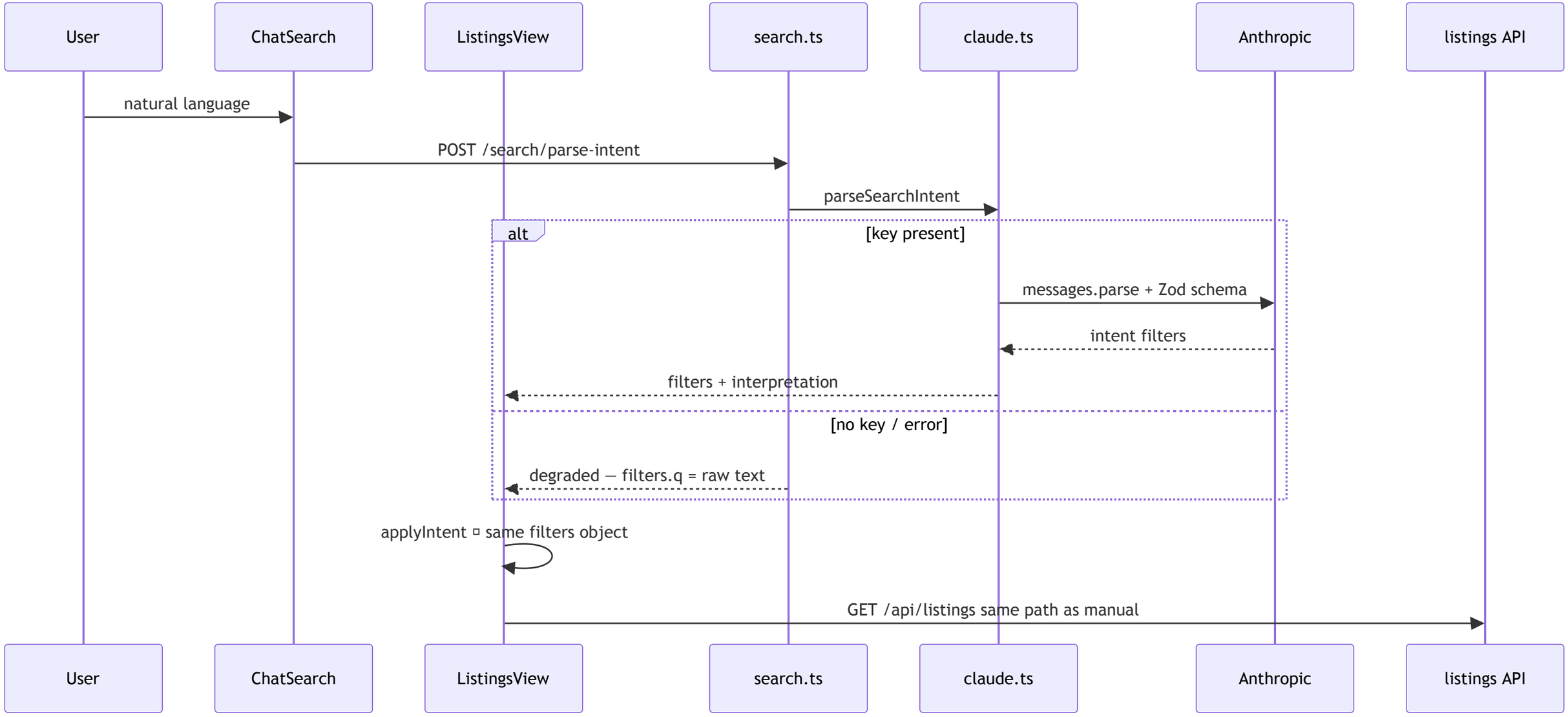









User
ListingsView
api/client
listings.ts
Zod
listingsRepo
MySQL
change filter / page
load — router.replace query
fetchListings filters
GET /api/listings?...
ListQuery.safeParse
parsed.data
findListings
SELECT + COUNT parameterized
items + total
User
ListingsView
api/client
listings.ts
Zod
listingsRepo
MySQL


User
ChatSearch
ListingsView
search.ts
claude.ts
Anthropic
listings API
natural language
POST /search/parse-intent
parseSearchIntent
alt
[key present]
messages.parse + Zod schema
intent filters
filters + interpretation
[no key / error]
degraded — filters.q = raw text
applyIntent ▫ same filters object
GET /api/listings same path as manual
User
ChatSearch
ListingsView
search.ts
claude.ts
Anthropic
listings API

